

Results, analysis, and design document 22.5.2026

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Group D

Scope: output from full run, input for report

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Objective:

This is our internal Group D working document. At this stage, we moved from several full runs of the track1 with modeling and design choices defined. The objective is to be transparent on design options and decisions, what are the alternatives considered and why discarded.

This document is not part of the formal submission. It is mainly to help us understand what has been set up and done so far, what are the conclusions Ariel took, and open for feedback.

No time invested in formatting and other formalities, more than communicate and document internally.

We may include part of this document -once final- in the submission report as annex/appendix in order to reflect the necessary for reproducibility and transparency.

This document follows the run of the 5 scripts aligned/homologous to the CRISP-DM workflow: how the different aspects and stages are connected, alongside the code execution; 3 tier process for design decisions.

Sections:

- 1) 3 Tier design process choices: feature set, cluster design and regression design.
- 2) Results, analysis and conclusions
- 3) References

1) 3 Tier design process choices

We organised our decisions using a three-tier framework based on how strong the justification is.

Tier 1 is the stronges

This is when a decision is anchored to a peer-reviewed academic source or to something we were formally taught in the module reading list or units. Foundational recognized text books like Bishop (2006), James et al. (2023) and Hastie, Tibshirani and Friedman (2009), or peer-reviewed papers like Rousseeuw (1987) were used.

Tier 2 is the middle tier

This is when we anchor a decision to documentation from the tools or libraries we use (like the scikit-learn user guide) or to a recognised practitioner source (like Brownlee from Machine Learning Mastery, or Patil on Towards Data Science). We call it standard practice.

Tier 3 is the least strong

This is when we make a decision based on practical considerations: the audience, the scope, time, use case, constraints, etc. We try to avoid tier 3 when possible unless is extremely necessary.

Feature set:

log1p(price):

is used because linear regression copes better with the type of raw price distribution present here, which has a high right skew and therefore needs variance stabilisation - which log1p provides - to improve the model fit (Bishop, 2006; James et al., 2023). Tier1

Integer mapping:

For regression, room_type and neighbourhood_group were encoded with simple integer mapping so they could be used by the scikit-learn regression models. This was a simple first-pass encoding choice for low-cardinality categories. Its limitation is that it can impose an artificial order on categories, the alternative and likely fix is one-hot encoding which is noted for “future work” sensitivity check (James et al., 2023; Hastie, Tibshirani and Friedman, 2009).

Tier 1

One-hot and drop first:

for clustering, room_type was one-hot encoded because K-Means measures distances between listings. A numeric room-type code would make the algorithm treat room types as an ordered number line, which is not correct. The final notebook uses drop_first=True, with Entire home/apt as the reference category. This avoids ordinal coding. A future work version could drop_first=False for sensitivity check for K-Means (Bishop, 2006; Hastie, Tibshirani and Friedman, 2009; James et al., 2023) Tier 1

StandardScaler (z-score):

By the way K-Means uses the distance between listings and this could produce that features with bigger ranges such as price would have imposed their domination in the distance calculation over features with smaller ranges like reviews per month, causing some distortion. This makes the cluster shapes weird. So then we apply standardisation by transforming the full feature matrix using StandardScaler (z-score), which puts every feature on the same scale. Scaling like this is the standard practice for clustering that relies on distances (Hastie, Tibshirani and Friedman, 2009; James et al., 2023). Tier 1

Regression design:

80/20 held-out test:

Typical proportion of 80% for training and 20% for testing to test whether our model can do price prediction on listings has never seen. Understand that the 80/20 is within the conventional range discussed in the treatment of resampling for datasets like this case (James et al., 2023). Tier 1 and Tier 2.

Three nested models architecture:

To understand the trade-off between complexity and predictive power of the model, we built upon what is taught in module Units 3 and 4 and applied a three-model progression: a simple linear regression on one feature, then a multiple linear regression on nine features (scaled), and finally a polynomial degree 2 on the scaled features. This way we can see how R² climbs and then levels off as we go from simpler to more complex models (Bishop, 2006; Hastie, Tibshirani and Friedman, 2009; James et al., 2023). What we tested: each model fit on the same training set and scored on the held-out 20% with R², MAE, MSE and RMSE, so the three models are compared on the same test set. Tier 1

Baseline model (linear regression on room_type_code only) and Multiple linear regression with all features:

To establish a baseline model and R2 outcome that everything else can be compared against, we picked `room_type_code` because it has the strongest correlation with `log_price` in the training set (absolute correlation 0.61), so thus we ran a linear regression model with this single feature. This way we have a strong baseline for the more complex models to compare with, pretty much the base way of setting up a regression baseline (Hastie, Tibshirani and Friedman, 2009; James et al., 2023). Tier 1 and Tier 2.

Now that we did the single-feature linear regression with the 80/20 hold-out test and obtained an R2, we move to next model. Then we ran a multiple linear regression, configured to use all nine numeric features. Like before, we used `StandardScaler` to make all the features comparable and standardised. And we obtained an R2 again. This is the typical progression that comes after a single-feature linear regression (Hastie, Tibshirani and Friedman, 2009; James et al., 2023). Tier 1 and Tier 2.

Polynomial degree:

Now that we did the multiple linear regression with the 80/20 hold-out test and obtained an R2, we move to the next model. Then we ran a polynomial regression of degree 2 on the same nine scaled features. Degree 2 was used to test simple curved patterns and paired relationships between the input features, such as availability with review activity or minimum nights with availability. It doesn't use price as input because here `log_price` is the prediction target. At the same time, this makes it still understandable at this level. Like before, we used the 80/20 hold-out test. And we obtained an R2 again. This is the typical progression that comes after multiple linear regression, to test if simple non-linear patterns or interactions add anything to the R2 (Hastie, Tibshirani and Friedman, 2009; James et al., 2023). Tier 1 and Tier 2.

Evaluation metrics: R2 and RMSE primary, MAE and MSE supporting

Now that we have run the three regression models and have an R2 for each, we needed a way to compare them with a reliable and academically defensible process, and being able to actually understand and judge how they perform. So we used four standard metrics.

R2 tells us how much of the `log_price` variation each model can explain (closer to 1 is better, 0 means the model is no better than just guessing the average).

RMSE tells us the typical size of the prediction error in the same units as our target `log_price` (lower is better). We use these two as the primary metrics.

We also added MAE (the average absolute error) and MSE (the average squared error) as supporting checks, because together they give a more complete picture of fit quality and error magnitude. This is pretty much the usual combination for evaluating regression models that is taught (Hastie, Tibshirani and Friedman, 2009; James et al., 2023). Tier 1 and Tier 2.

Multicollinearity diagnostic (VIF on the full feature):

Going back a step, when we ran the multiple linear regression with all nine features, we needed to check that none of those features were too correlated with each other. If they are, the coefficients become unstable and hard to interpret. So we used VIF (Variance Inflation Factor) on the full feature matrix, because it is what I learn to use in diagnostic for this situations. A common threshold is 5, anything above means there is a multicollinearity problem worth addressing. We computed VIF for all nine features and the values came out between 1.06 and 1.63, well below 5, so we did not need to remove any feature. We kept VIF as a reporting diagnostic (James et al., 2023). Tier 1 and Tier 2.

Clustering design:

Clustering method K-Means:

To group the NYC listings into segments that we can actually use for business decisions, we needed to pick a clustering algorithm and we went with K-Means. Rationals are varied:

is taught in module Unit 6 and it is also one of the most common clustering methods.

it produces hard partitions, which means each listing belongs to exactly one segment and we end up with clear groups for tiered host guidance this is under the assumption that the limitations of the dataset are like Ariel mentioned and therefore the best we can do is guidance and assistance but not much more

the centroids of each segment represents the average listing profile which gives the business some meaning

It scales well on a dataset of our size (about 48,800 listings);

and it is the usual choice for marketplace and customer segmentation in industry.

The alternatives I considered without testing: hierarchical clustering which discarded due to scaling at this size and the well known DBSCAN which I understand is more suitable for suited to spatial or density-based problems than our standardized tabular features. This we learned in previous modules and further literature.

K-Means then is chosen as the best fit for what we needed (Bishop, 2006; Hastie, Tibshirani and Friedman, 2009). Tier 1

K sweep 2 to 6:

Now that we picked K-Means, we needed to decide how many clusters to actually test. We swept K from 2 to 6 because this range is wide enough to actually see the elbow in inertia so we can pick a good K and narrow enough to keep the resulting segments easy to explain, in this case 3 was the right value to show segments that aligns almost entirely to the room type which fits the use case. This is the same range used in module Unit 6 seminar activity wheterAUS so we follow the same pattern and sustain with the cluster literature (Tippaya Thinsungnoen et al., 2015). Tier 1

K selection - primary (elbow on inertia):

To actually choose the K value, we used the elbow method on inertia (the within-cluster sum of squared errors). The idea is simple: as we add more clusters, inertia drops because each cluster covers less ground. Similar to complexity in regression results, here there is a moment where adding more clusters does not help anymore, which is the elbow. We picked this method because it is the taught way of choosing K for K-Means (Tippaya Thinsungnoen et al., 2015). Tier 1

K selection (silhouette score, sampled):

Like before with the elbow, we also wanted a supporting check on the K value, so we added the silhouette score. The silhouette tells us how well each listing fits into its assigned cluster compared to the nearest other cluster. We sampled it for runtime for simplicity. Although possible, we did not run it against the full dataset; based on the results it was not needed. We used it as a supporting signal rather than the primary one because in our sweep the silhouette kept rising as K grew, while the elbow gave us a clearer point to pick K (Rousseeuw, 1987). Tier 1

Final K chosen $K = 3$:

Now that we have both the elbow and the silhouette, we needed to choose the final K.

We went with $K = 3$ because the elbow between $K = 2$ and $K = 3$ marks the practical bending point on this dataset.

K = 3 also aligns well with the three room_type categories (entire home, private room, shared room), so the segmentation is both statistically supported by the elbow and structurally aligned with the main categorical feature in the dataset.

Although higher K values like K = 5 or 6 could have been used, were deliberate left them as future work, as considered not needed for the current decisioning process (Rousseeuw, 1987; Tippaya Thinsungnoen et al., 2015). Tier 1

2)Results, analysis and conclusions

Dataset limitations

This is a 2019 listing picture.

Big missings: bookings, revenue, occupancy by date, price to behaviour data, no guest records, search to booking connection.

Mid missings: other features such as pictures, description, what happens before and after publish and booking, ratings connection, etc.

Observational, not causal

From the get-go: not possible to do price engine, only grouping and price guidance for hosts based on peer.

EDA findings

Cleaned data: 48,895 -> 48,847 listings.

Price is skewed: median \$106, mean \$153, p99 \$799.

price contained good pricing info but is too right skewed due to a few high prices introducing distortions. To avoid those high prices affecting the regression modeling, I created log_price from the original price column using a logarithmic scale.

Manhattan + Brooklyn concentrates approx. 85% of listings.

35.8% with 0 availability, how to treat and what means ?

Regression run and results

Regression can predict in this case the observed price patterns, we cant do price engine.

Model path: room-type baseline -> multiple linear -> polynomial degree 2.

R2 improves: 0.3720 -> 0.4803 -> 0.5343.

Outcome and result: final model can explain approx. half of log-price pattern.

RMSE improves: 0.5475 -> 0.4981 -> 0.4714.

Outcome and result: prediction error gets smaller good !, however model has strong limitations still.

Regression results connected to business problem and outcome

Room type is the strongest pricing signal.

Location and group within neighborhood complements.

VIF max: 1.625.

Outcome and result: VIF is low, so features are not strongly duplicating each other. Still does not make regression causal so the price prediction limitation remains.

How to use? For hosts look at your competition price and benchmark against.

Clustering run and K choice

K-Means used to group similar listings.

Inputs log_price, minimum_nights, reviews_per_month, availability_365 and room-type dummies.

Scaling used because K-Means is based on distance, tested/used to see when and which number feature dominates the grouping and avoid distortions.

K checked from 2 to 6.

K = 3 chosen

Clustering results

Arrived to 3 listing grouping:

Entire: 52.6%, \$174, 113 days/year/availability.

Private: 45.1%, \$71, 110 days/year/availability.

Shared: 2.4%, \$54, 162 days/year/availability.

These are listing segments

There are no customer segments

Going back to Track 1 objectives, from assignment:

Track	Machine Learning Approach	Focus Areas
Track 1	Classical ML (Regression & Clustering)	Price prediction, demand trends, customer segmentation

- 1) Price prediction: expectation: price engine. Not a price engine due to dataset limitations, only observational by using list price patterns, the best regression model can explain 53% of the observed log_price variation.
- 2) Demand trends: expectation: demand patterns, seasonal, real demand, etc..
Possible ? not like this, indirect such as demand signals and hypothesize on zero availability ones. Once more, limitations due to dataset features and snapshot like.
Customer segmentation: expectation customer clustering. Possible ? no, as there are no true customer information in the dataset, conversion rate, nothing strongly

direct to understand what is and how behaves a customer. Other segmentation like listings possible and that's what was analysed.

Conclusion overall: All in all, it feels like the focus areas where meant for a different dataset, OR it is expected that the areas cant be answered directly but indirectly with best alternative and stating limitations , show critical thinking, etc.

Recommendations

For hosts: maybe provide hosts with peer comparison for better informed decisions on pricing

Promote and manage discount (or not) based on knowns: cluster result price and availability, room type concentration and eda findings geo concentration.

For Airbnb, expand other business geo areas ? too concentrated because .. need to hypothesize or common use case behaviour... room type expand or invest ? Evaluate this in conjunction with business strategies not numbers alone, blue ocean, red ocean, just as examples what would apply, others. Objective: more hosts, more revenue, more profit.

Peer-price benchmark pilot: because regression explains about half of the observed log_price pattern, Airbnb can test host guidance such as "similar listings suggest this range." Potential benefit: better host pricing to increase guest acceptance

Manhattan Brooklyn concentration red-ocean strategy: Manhattan and Brooklyn concentrates 85% of listings, it could be tested A) what happens if on dense geo the prices are increased ?

- and B) test blue ocean in Queens, Bronx and Staten Island to understand if this is because there is no demand OR no hosts, and if the latter then recruit more hosts to expand the addressable market beyond concentrated areas.

SWOT on the 35.8% of listings show zero availability to understand whether those listings are booked, use by their owners, or maintenance or whateve reason and then once known, decrease the zero availability, maybe even support the host or incentivise with marketing campaings.

Market segmentation: for Airbnb using the clustering results 3 listing segments, test customised messages based on cluster and room type.

To close

Big limitations: not possible price engine/causal pricing, full market understanding, customer behaviour and booking, bookings, true customer segmentation.

Reason: dataset is a listing snapshot, not booking/revenue history.

Big yes: very useful decision support for hosts, provide more transparent host information, speculative improvements for Airbnb through assumptions through business strategies and hypothesis testing

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